Questions

Q1.

In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a, where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

RocketRevise

Given that the **total** number of seats in the first 10 rows is 360

(a) find the value of a.

(2)

Given also that the total number of seats in the n rows is 2146

(b) show that

$$n^2 + 8n - 1073 = 0$$

(2)

(c) Hence

(i) state the number of rows of seats in the theatre,

(ii) find the maximum number of seats in any one row.

(3)

www.RocketRevise.com

(Total for question = 7 marks)

(Q08 WMA12/01, Oct 2023)

Q2.

compiled by Ahmed Gul www.RocketRevise.com

A company that owned a silver mine

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

RocketRevise

Given that this occurred at the end of year N,

(b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

(c) Hence, state the value of N.

(1)

(Total for question = 6 marks)

(**Q05 WMA12/01, Oct 2021**)

Q3.

Ben is saving for the deposit for a house over a period of 60 months.

www.RocketRevise.com

Ben saves £100 in the first month and in each subsequent month, he saves £5 more than the previous month, so that he saves £105 in the second month, £110 in the third month, and so on, forming an arithmetic sequence.

(a) Find the amount Ben saves in the 40th month.

(2)

(b) Find the total amount Ben saves over the 60-month period.

(3)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Lina is also saving for a deposit for a house.

Lina saves £600 in the first month and in each subsequent month, she saves £10 less than the previous month, so that she saves £590 in the second month, £580 in the third month, and so on, forming an arithmetic sequence.

Given that, after n months, Lina will have saved exactly £18 200 for her deposit,

(c) form an equation in n and show that it can be written as

$$n^2 - 121n + 3640 = 0$$

(3)

(d) Solve the equation in part (c).

RocketRevise

(2)

(e) State, with a reason, which of the solutions to the equation in part (c) is **not** a sensible value for n.

(1)

(Total for question = 11 marks)

(**Q05 WMA12/01, Oct 2020**)

Q4.

An arithmetic series has first term a and common difference d.

(a) Prove that the sum of the first n terms of the series is

$$\tfrac{1}{2}n[2a + (n-1)d]$$

www.RocketRevise.com

(4)

A company, which is making 200 mobile phones each week, plans to increase its production.

The number of mobile phones produced is to be increased by 20 each week from 200 in week 1 to 220 in week 2, to 240 in week 3 and so on, until it is producing 600 in week N.

(b) Find the value of N

(2)

The company then plans to continue to make 600 mobile phones each week.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(c) Find the total number of mobile phones that will be made in the first 52 weeks starting from and including week 1.

(5)

(Total for question = 11 marks)

(**Q05 WMA12/01, Specimen papers**)

RocketRevise

Q5.

(i) An arithmetic series has first term α and common difference d.

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\}$$

(3)

(ii) A sequence u_1, u_2, u_3,... is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5

(1)

(b) $\sum_{n=1}^{59} u_n$

www.RocketRevise.com

(3)

(Total for question = 7 marks)

(**Q08 WMA12/01, Jan 2020**)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q6.

A geometric sequence has first term a and common ratio r, where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

RocketRevise

(b) Hence find the value of a.

(2)

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n.

(*Solutions based entirely on graphical or numerical methods are not acceptable.*)

(4)

(Total for question = 7 marks)

(**Q07 WMA12/01, Jan 2023**)

Q7.

Given $\log_a b = k$, find, in simplest form in terms of k,

www.RocketRevise.com

(i) $\log_a\left(\frac{\sqrt{a}}{b}\right)$

(2)

(ii) $\frac{\log_a a^2 b}{\log_a b^3}$

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(2)

(iii) $\sum_{n=1}^{50}(k+\log_a b^n)$

(3)

(Total for question = 7 marks)

(**Q07 WMA12/01, Oct 2019**)

Q8.

Kim starts working for a company.

- In year 1 her annual salary will be £16 200
- In year 10 her annual salary is predicted to be £31 500

Model *A* assumes that her annual salary will increase by the same amount each year.

(a) According to model *A*, determine Kim's annual salary in year 2.

(3)

Model *B* assumes that her annual salary will increase by the same percentage each year.

(b) According to model *B*, determine Kim's annual salary in year 2. Give your answer to the nearest £10

(3)

(c) Calculate, according to the two models, the difference between the total amounts that Kim is predicted to earn from year 1 to year 10 inclusive. Give your answer to the nearest £10

(3)

(Total for question = 9 marks)

(**Q07 WMA12/01, June 2019**)

www.RocketRevise.com

Q9.

In a geometric sequence u_1, u_2, u_3, ...

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0$$

(3)

Given that the geometric sequence has a sum to infinity,

(b) find u_1

(3)

(c) find S_∞

(2)

(Total for question = 8 marks)

(**Q06 WMA12/01, June 2022**)

Q10.

A geometric sequence has first term a and common ratio r

Given that $S_\infty = 3a$

(a) show that $r = \frac{2}{3}$

(2)

Given also that

compiled by Ahmed Gul www.RocketRevise.com

$u_2 - u_4 = 16$

where u^k is the kth term of this sequence,

(b) find the value of S_{10} giving your answer to one decimal place.

(5)

(Total for question = 7 marks)

(**Q08 WMA12/01, Oct 2022**)

RocketRevise

Q11.

The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

(a) Find, according to the model, the adult population of the town at the start of 2032

(3)

It is also modelled that every member of the adult population gives £5 to local charity at the start of each year.

(b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000

(3)

(Total for question = 6 marks)

www.RocketRevise.com

(**Q02 WMA12/01, Oct 2019**)

Q12.

The first term of a geometric series is 20 and the common ratio is

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

$\frac{7}{8}$. The sum to infinity of the series is S_∞

(a) Find the value of S_∞

(2)

The sum to N terms of the series is S_N

(b) Find, to 1 decimal place, the value of S_{12}

(2)

(c) Find the smallest value of N, for which $S_\infty - S_N < 0.5$

RocketRevise

(4)

(Total for question = 8 marks)

(**Q02 WMA12/01, Specimen papers**)

Q13.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$2k - 15$ k $k + 4$

www.RocketRevise.com

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$

(2)

(b) predict the number of downloads in the 4th month.

(4)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the *N*th month.

(c) Calculate the value of *N* according to the model.

(Total for question = 6 marks)

(**Q06 WMA12/01, June 2023**)

RocketRevise

Q14.

A geometric series has first term *a* and common ratio *r*.

(a) Prove that the sum of the first *n* terms of this series is given by

$$S_n = \frac{a(1-r^n)}{1-r}$$

(3)

The second term of a geometric series is -320 and the fifth term is $\frac{512}{25}$

(b) Find the value of the common ratio.

(2)

(c) Hence find the sum of the first 13 terms of the series, giving your answer to 2 decimal places.

(3)

www.RocketRevise.com

(Total for question = 8 marks)

(**Q08 WMA12/01, Oct 2020**)

Q15.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

(a) Calculate the distance he will run on Sunday in week 5 under training plan *A*.

RocketRevise

(3)

In training plan *B*, he will increase the distance he runs each Sunday by the same percentage.

(b) Calculate the distance he will run on Sunday in week 5 under training plan *B*. Give your answer in km to one decimal place.

(3)

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
- the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km

(c) (i) find the maximum value of x, if he uses training plan *A*,

(ii) find the maximum value of x, if he uses training plan *B*.

(5)

(Total for question = 11 marks)

www.RocketRevise.com

(**Q10 WMA12/01, Jan 2021**)

Q16.

Adina is saving money to buy a new computer. She saves Â£5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

(a) (i) state with a reason whether an arithmetic series or a geometric series should be used,

(ii) write down an expression, in terms of n, for the amount, in pounds (£), saved in week n.

(3)

Given that the computer Adina wants to buy costs £350

(b) find the number of weeks it will take for Adina to save enough money to buy the computer.

RocketRevise

(4)

(Total for question = 7 marks)

(**Q01 WMA12/01, June 2021**)

Q17.

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by p % each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

www.RocketRevise.com

(a) find the value of p, giving your answer to 2 decimal places.

(3)

According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of N.

(5)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(Total for question = 8 marks)

(Q05 WMA12/01, Jan 2020)

Q18.

(i) A geometric sequence has first term 4 and common ratio 6

Given that the nth term is greater than 10^{100}, find the minimum possible value of n.

RocketRevise

(3)

(ii) A different geometric sequence has first term a and common ratio r.

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

(a) show that

$$25r^2 - 25r - 6 = 0$$

(3)

(b) Write down the solutions of

$$25r^2 - 25r - 6 = 0$$

(1)

Hence,

(c) state the value of r, giving a reason for your answer,

www.RocketRevise.com

(1)

(d) find the sum of the first 4 terms of the series.

(2)

(Total for question = 10 marks)

www.RocketRevise.com

RocketRevise

www.RocketRevise.com

www.RocketRevise.com